

David B. Hamilton
Vice President

440-280-5382

October 04, 2017
L-17-305

10CFR50.73(a)(2)(v)(C)
10CFR50.73(a)(2)(v)(D)

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

SUBJECT:
Perry Nuclear Power Plant
Docket No. 50-440, License No. NPF-58
Licensee Event Report Submittal

Enclosed is Licensee Event Report (LER) 2017-005, "Controller Malfunction results in Momentary Degradation of Secondary Containment Pressure". There are no regulatory commitments contained in this submittal.

If there are any questions or if additional information is required, please contact Mr. Nicola Conicella, Manager – Regulatory Compliance, at (440) 280-5415.

Sincerely,



David B. Hamilton
Vice President

Enclosure:
LER 2017-005

cc: NRC Project Manager
NRC Resident Inspector
NRC Region III

Enclosure
L-17-305

LER 2017-005

**LICENSEE EVENT REPORT (LER)**
(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

Perry Nuclear Power Plant

2. DOCKET NUMBER

05000-440

3. PAGE

1 OF 3

4. TITLE:

Controller Malfunction results in Momentary Degradation of Secondary Containment Pressure

5. EVENT DATE

MONTH	DAY	YEAR
08	14	2017

6. LER NUMBER

YEAR	SEQUENTIAL NUMBER	REV NO.
2017	005	00

7. REPORT DATE

MONTH	DAY	YEAR
10	04	2017

8. OTHER FACILITIES INVOLVED

FACILITY NAME	DOCKET NUMBER
	05000
FACILITY NAME	DOCKET NUMBER
	05000

9. OPERATING MODE

1

11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A

100

12. LICENSEE CONTACT FOR THIS LER**LICENSEE CONTACT:**

George Dujanovic – Regulatory Compliance

TELEPHONE NUMBER (Include Area Code)

440-280-5200

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANUFACTURER	REPORTABLE TO EPIX
B	VC	PC	B040	Y					

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On August 14, 2017 at 2257, with both trains of the Annulus Exhaust Gas Treatment System (AEGTS) running in parallel, Secondary Containment vacuum momentarily degraded to 0.52" water gauge when AEGTS B was shutdown to Standby Readiness. After an approximately 15 second delay, AEGTS A responded to maintain the proper vacuum. Technical Specification (TS) Surveillance Requirement (SR) 3.6.4.1.1 requires a minimum of 0.66" water gauge differential pressure to be maintained between ambient and annulus pressure. The failure to comply with SR 3.6.4.1.1 caused Secondary Containment to be inoperable resulting in Technical Specification 3.6.4.1 not being met.

The cause was determined to be equipment age-related degradation of the AEGTS A controller circuit card, which was subsequently replaced. There were no other event-related equipment malfunctions.

Since AEGTS is not a core damage mitigation system and is not credited in the PRA model to mitigate large and early containment releases, the inoperability of the AEGTS is determined to be of small safety significance. This event is being reported in accordance with 10CFR50.73(a)(2)(v)(C) and 10CFR50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of a safety function.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Perry Nuclear Power Plant Unit 1	05000-440	2017	- 005	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

INTRODUCTION

On August 14, 2017 at 2257, with both trains of the Annulus Exhaust Gas Treatment System [VC] (AEGTS) running in parallel, Secondary Containment [NH] vacuum momentarily degraded to 0.52" water gauge differential pressure when AEGTS B was shutdown to Standby Readiness. After an approximate 15 second delay, AEGTS A responded to maintain the normal vacuum. Technical Specification (TS) Surveillance Requirement (SR) 3.6.4.1.1 requires a minimum of 0.66" water gauge differential pressure to be maintained between ambient and annulus pressure at all times. The failure to comply with SR 3.6.4.1.1 caused Secondary Containment to be inoperable resulting in Technical Specification 3.6.4.1 not being met.

EVENT DESCRIPTION

On August 14, 2017 at 2019 hours, with the plant in mode 1 and at 100% power, AEGTS A was manually started in parallel with AEGTS B in support of flow switch calibration testing.

At 2257 hours, AEGTS B was shutdown to standby. During the shutdown, Operations personnel received an unexpected Annulus Differential Pressure Low alarm. Secondary Containment vacuum dropped to 0.52" water gauge for approximately 15 seconds. After the approximate 15 second delay, AEGTS A responded to maintain the normal vacuum. Technical Specification (TS) 3.6.4.1 "Secondary Containment" condition A was entered. Condition A addresses secondary containment being inoperable in modes 1, 2, or 3, with operability to be restored within 4 hours. Surveillance Requirement (SR) 3.6.4.1.1 was not met which states, "Verify secondary containment vacuum is $\geq$ 0.66 inch of vacuum water gauge".

On August 15, 2017 at 0458 hours, event notification EN# 52903 was made to the NRC Operations Center in accordance with 50.72(b)(3)(v)(C) & 50.72(b)(3)(v)(D) for loss of safety function.

CAUSE

A causal analysis was conducted and determined the most likely cause to be age-related degradation of the Bailey [B040] controller card [PC] transformer nylon screw, resulting in less than optimum response.

EVENT ANALYSIS

With respect to the degraded controller and the resulting sluggish response of the AEGTS A subsystem, the Loss of Coolant Accident (LOCA) analysis assumes the bypassing of the Secondary Containment for the first 40 seconds of the LOCA. Additionally, since the gap release (radiological release due to core damage) does not occur until 120 seconds, the capability of the AEGTS A subsystem has been shown to support recovery of the negative pressure in the annulus to greater than the required minimum of 0.66 inches vacuum water gauge within the required analytical assumptions. The LOCA is the only Design Basis Accident (DBA) for which secondary containment is credited and therefore, the recovery within two minutes will support the LOCA analysis. There is reasonable assurance that Secondary Containment can perform the required design and license functions. As a result, this condition was not a safety system functional failure.

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NARRATIVE

From the Probabilistic Risk Assessment perspective, AEGTS is not a core damage mitigation system and is not credited in any PRA models in mitigating large and early containment releases. Additionally, from an availability / functionality perspective, AEGTS was fully capable of performing its' function. On this basis, the inoperability of the AEGTS for the period of 15 seconds is determined to be of small risk (safety) significance.

This event is being reported in accordance with 10CFR50.73(a)(2)(v)(C) and 10CFR50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfilment of a safety function.

CORRECTIVE ACTIONS

Corrective actions in response to the event include: closure of the Prompt Operability Determination (POD) following replacement of the Bailey controller circuit card (completed), revise the Preventative Maintenance (PM) plan for 1M15R0060A/B-1 to replace the transformer nylon screw every six years versus as required, and create PM tasks for other same-type controllers to change the transformer nylon screw every six years.

PREVIOUS SIMILAR EVENTS

A review of LERs and the Corrective Action database for the past three years identified no similar events.

COMMITMENTS

There are no regulatory commitments contained in this report. Actions described in this document represent intended or planned actions, are described for the NRC's information, and are not regulatory commitments.